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Hereditary and Acquired Ichthyosis Vulgaris

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Continuing Education Activity

Hereditary and acquired **ichthyosis** vulgaris are cutaneous disorders of keratinization that manifest clinically by a widespread scaling of the skin. **Ichthyosis** vulgaris is associated with loss of function FLG gene mutations, which leads to a defective skin barrier. This activity reviews the evaluation and treatment of hereditary and acquired **ichthyosis** vulgaris and highlights the role of the interprofessional team in the care of patients with this condition.

Objectives:

- Identify the etiology of hereditary and acquired **ichthyosis** vulgaris.
- Describe the appropriate evaluation of hereditary and acquired **ichthyosis** vulgaris.
- Outline the management options available for hereditary and acquired **ichthyosis** vulgaris.
- Summarize interprofessional team strategies for improving care coordination and communication to advance the care of hereditary and acquired **ichthyosis** vulgaris and improve outcomes.

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Introduction

The term "**ichthyosis**" implies a generalized scaly dermatosis, without the component of erythema. The ichthyoses are disorders of cornification and are classified as inherited or acquired. Hereditary **ichthyosis** is further classified as syndromic and non-syndromic. **Ichthyosis** vulgaris (IV) is an inherited form of non-syndromic **ichthyosis**, with a very early age of onset. **Ichthyosis** vulgaris is associated with loss of function FLG gene mutations, which leads to a defective skin barrier. Historically, the first case of **ichthyosis** vulgaris was described in 1806 as "ichthyose nacrée." Willan introduced the term "**ichthyosis**," derived from the Greek root "ichthy," meaning fish.[1][2]

Etiology

Hereditary **ichthyosis** (HI) is an autosomal dominant genetic disorder that develops in early childhood. Whereas, acquired **ichthyosis** (AI) may be associated with underlying cancer, such as Hodgkin's disease, multiple myeloma,[3] cutaneous T cell lymphoma or with an underlying systemic disorder such as systemic lupus erythematosus,[4] acquired immunodeficiency syndrome (AIDS) or health conditions that reduce lipid and vitamin absorption including celiac disease and Crohn's disease. Certain medications have been reported to induce AI, such as cholesterol-lowering agents, allopurinol, targeted cancer therapy (EGFR inhibitors, BRAF inhibitors), and many more.

Epidemiology

IV is the most common **ichthyosis**, with an estimated prevalence of 1 in 100 to 250. The mode of inheritance is autosomal semi-dominant with a milder phenotype in heterozygous individuals. The overall penetrance is 80%-95%. The age of onset is in infancy, with the majority of patients having clearly characteristic manifestations by 5 years of age. Sex or ethnic predilection has not been observed.

Pathophysiology

The central event in the pathogenesis of **ichthyosis** is the disruption of the epidermal barrier. The integrity of the cornified cell envelope is responsible for the maintenance of the barrier function of the skin. In **ichthyosis** vulgaris, loss of function mutations of the FLG gene leads to defective filaggrin formation, which is an important component of the cornified cell envelope.

The stratum granulosum is characterized by keratohyaline granules. Profilaggrin is an important component of these granules. During keratinocyte differentiation, profilaggrin is cleaved into filaggrin peptides, which in turn, are responsible for the alignment of keratin intermediate filaments. Ultimately, these complexes get cross-linked and are responsible for the compactness of the cell envelope.

Loss of function mutations of FLG, commonly, the R501X and c.2282del4, lead to an absolute or relative deficiency of filaggrin.[5][6] This causes a disruption of the mechanical barrier, leading to excessive transepidermal water loss. Moreover, it also disrupts the immunological barrier of the skin by increasing the propensity of penetration of allergens and microorganisms and the resultant cutaneous inflammatory responses. Hence, patients of **ichthyosis** vulgaris have a higher tendency of atopic dermatitis, as well as hand eczema, irritant contact dermatitis, and allergic contact dermatitis.[7][8][9][10]

In normal epidermis, filaggrin is eventually degraded into water-retaining amino acids, in particular histidine, that serves as a natural moisturizer. FLG mutations cause a deficiency of histidine and its metabolites, one of which is urocanic acid, which is a natural photoprotective agent. Hence in **ichthyosis** vulgaris, there is an upregulation in UVB induced synthesis of vitamin D, ultimately leading to an increase in serum vitamin D levels.[11][12]

Histopathology

The epidermis shows a mild orthokeratotic hyperkeratosis with or without hypo/agranulosis. There is an absence of inflammatory infiltrate.

Electron microscopy is diagnostic, which shows absent or structurally abnormal keratohyaline granules.

History and Physical

The spectrum of clinical features of **ichthyosis** vulgaris varies from mild xerosis to large adherent scales (lizard skin). Heterozygous FLG mutations usually present in infancy or early childhood. Fine, whitish, scales with predominant extensor distribution is the hallmark manifestation. The flexural areas are typically spared, due to the presence of humidity in these areas. In homozygous individuals, the manifestations are much more severe with large, centrally adherent scales, especially on the lower limbs and trunk and palmoplantar fissuring and hyperkeratosis. Severe cases may show facial involvement with mild ectropion and perioral fissuring. The scalp shows extensive scaling. Nails and mucosa are usually unaffected.

The scaling shows a seasonal exacerbation in winter and is relatively improved in the summer and in humid climates. Even though it persists lifelong, **ichthyosis** vulgaris tends to improve with age.

About 25% to 50% of patients of **ichthyosis** vulgaris show associated atopic dermatitis, asthma, and hay fever, with filaggrin deficiency being the major predisposing factor. Keratosis pilaris and palmoplantar hyperlinearity are other associated features.

Evaluation

A high suspicion for **ichthyosis** vulgaris should be included in the evaluation of patients consulting for skin disorders. A family history of a similar dermatosis and a thorough inspection of the skin should be obtained. The diagnosis of **ichthyosis** vulgaris is based on several factors. These factors include the time of onset of clinical symptoms, the skin phenotype (color, scale pattern, the presence of erythroderma, a collodion membrane at birth), associated cutaneous manifestations, family history and hereditary transmission, histopathologic findings, absence or presence of extracutaneous manifestations, and genetic testing. Genetic testing may be beneficial for the confirmation of diagnosis and for genetic counseling.

Treatment / Management

The main treatment approach of both HI and AI includes hydration of the skin and application of an ointment to prevent evaporation. Urea-based emulsion, composed of 10% urea, natural moisturizing factors, and ceramides used twice daily, represents an effective therapeutic option.[13] Topical retinoids and alpha-hydroxy acids may be helpful for some patients. Topical retinoids act by reducing cohesiveness of epithelial cells, stimulating mitosis and turnover, and suppressing keratin synthesis.[14] Mild topical steroids may be useful in the case of pruritus. Treatment of the underlying systemic condition may improve the AI.

Differential Diagnosis

Differential diagnoses of **ichthyosis** vulgaris include:

- Allergic contact dermatitis
- Eczema
- Drug eruptions
- Harlequin **ichthyosis**
- Irritant contact dermatitis
- Lamellar **ichthyosis**
- X-linked **ichthyosis**

Prognosis

HI has an excellent prognosis as symptoms usually improve with age. The prognosis of AI depends on the severity of the underlying systemic disease or cancer.

Complications

- Pruritus - excessive pruritus leads to lichenification of the skin
- Atopic dermatitis, hay fever, asthma
- Secondary bacterial infections
- Interference with sweating, due to excessive scaling
- In severe cases: dehydration, nutritional deficiency (loss of proteins and minerals from scales)

Deterrence and Patient Education

As a conclusion of the study, collective therapeutic education programs seem to be very interesting. In general, the contributions of a therapeutic education program to the management of genodermatoses can help in building trust between patients, their families, and carers and promote the encouragement of self-expression and the autonomy of patients and their families.

Enhancing Healthcare Team Outcomes

Ichthyosis has a significant psychological impact on each patient, as almost all visible skin surfaces are affected. This frequently leads to social rejection. The management of these patients is difficult, and the treatment is expensive and time-consuming. Collaboration between primary clinicians, dermatologists, psychiatrists, nurses, and social workers is recommended.

Review Questions

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References

1. Takeichi T, Akiyama M. Inherited **ichthyosis**: Non-syndromic forms. J Dermatol. 2016 Mar;43(3):242-51. [PubMed: 26945532]
2. Thyssen JP, Kezic S. Causes of epidermal filaggrin reduction and their role in the pathogenesis of atopic dermatitis. J Allergy Clin Immunol. 2014 Oct;134(4):792-9. [PubMed: 25065719]
3. Dar NR, Raza N, Khan A, Amin MU. Paraneoplastic Addisonian pigmentation and acquired **ichthyosis** as presenting features of multiple myeloma. J Coll Physicians Surg Pak. 2011 Jan;21(1):40-2. [PubMed: 21276385]
4. Biver-Dalle C, Gil H, Méaux-Ruault N, Mermet-Ginet I, Aubin F, Humbert P, Magy-Bertrand N. [Acquired **ichthyosis** and livedoid palmoplantar keratoderma: two unusual skin manifestations of systemic lupus erythematosus]. Rev Med Interne. 2012 Mar;33(3):e14-6. [PubMed: 21429634]
5. Oji V, Seller N, Sandilands A, Gruber R, Gerss J, Hüffmeier U, Hamm H, Emmert S, Aufenvenne K, Metze D, Luger T, Loser K, Hausser I, Traupe H, McLean WH. **Ichthyosis** vulgaris: novel FLG mutations in the German population and high presence of CD1a+ cells in the epidermis of the atopic subgroup. Br J Dermatol. 2009 Apr;160(4):771-81. [PubMed: 19183181]
6. Gruber R, Janecke AR, Fauth C, Utermann G, Fritsch PO, Schmuth M. Filaggrin mutations p.R501X and c.2282del4 in **ichthyosis** vulgaris. Eur J Hum Genet. 2007 Feb;15(2):179-84. [PubMed: 17164798]
7. Hanifin JM. Evolving concepts of pathogenesis in atopic dermatitis and other eczemas. J Invest Dermatol. 2009 Feb;129(2):320-2. [PubMed: 18719604]
8. Winge MC, Hoppe T, Berne B, Vahlquist A, Nordenskjöld M, Bradley M, Törmä H. Filaggrin genotype determines functional and molecular alterations in skin of patients with atopic dermatitis and **ichthyosis** vulgaris. PLoS One. 2011;6(12):e28254. [PMC free article: PMC3229525] [PubMed: 22164253]
9. Kang TW, Lee JS, Oh SW, Kim SC. Filaggrin mutation c.3321delA in a Korean patient with **ichthyosis** vulgaris and atopic dermatitis. Dermatology. 2009;218(2):186-7. [PubMed: 18841000]
10. Xiong Z, Luo S, Xu X, Zhang L, Peng H, Li W, Xue J, Chen X, Hu Z, Xia K. Novel FLG mutations associated with **ichthyosis** vulgaris in the Chinese population. Clin Exp Dermatol. 2012 Mar;37(2):177-80. [PubMed: 22299762]
11. Zhong W, Cui B, Zhang Y, Jiang H, Wei S, Bu L, Zhao G, Hu L, Kong X. Linkage analysis suggests a locus of **ichthyosis** vulgaris on 1q22. J Hum Genet. 2003;48(7):390-2. [PubMed: 12838398]
12.

Liu P, Yang Q, Wang X, Feng A, Yang T, Yang R, Wang P, Yuang M, Liu M, Liu JY, Wang QK. Identification of a genetic locus for **ichthyosis** vulgaris on chromosome 10q22.3-q24.2. J Invest Dermatol. 2008 Jun;128(6):1418-22. [PubMed: 18079749]

13. Benintende C, Boscaglia S, Dinotta F, Lacarrubba F, Micali G. Treatment of **ichthyosis** vulgaris with a urea-based emulsion: videodermatoscopy and confocal microscopy evaluation. G Ital Dermatol Venereol. 2017 Dec;152(6):555-559. [PubMed: 29050444]
14. Haas AA, Arndt KA. Selected therapeutic applications of topical tretinoin. J Am Acad Dermatol. 1986 Oct;15(4 Pt 2):870-7. [PubMed: 3534024]

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